Network Testbed Sprint 1

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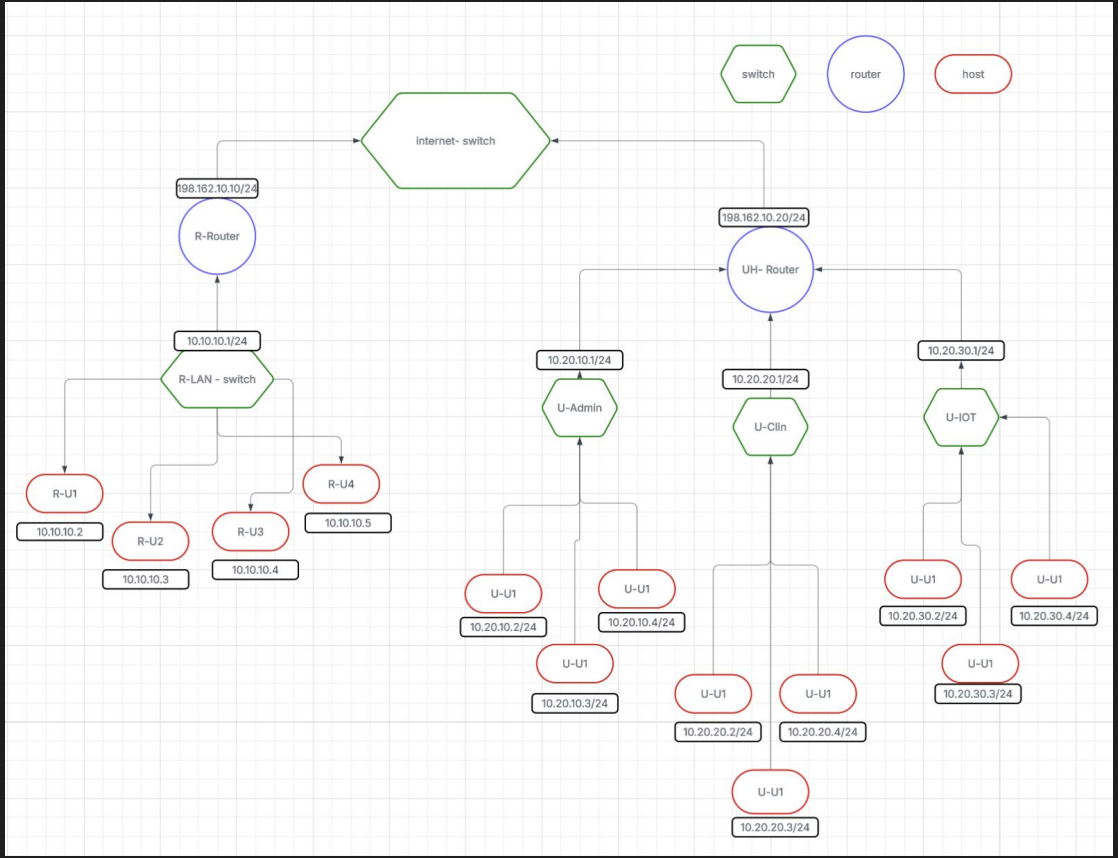
Project Overview

A network simulation platform for emulating network topologies and simulating network traffic during cyber attack scenarios. Its main use will be in testing network protocols. Performance will be measured based on simulated network conditions and cyber attacks

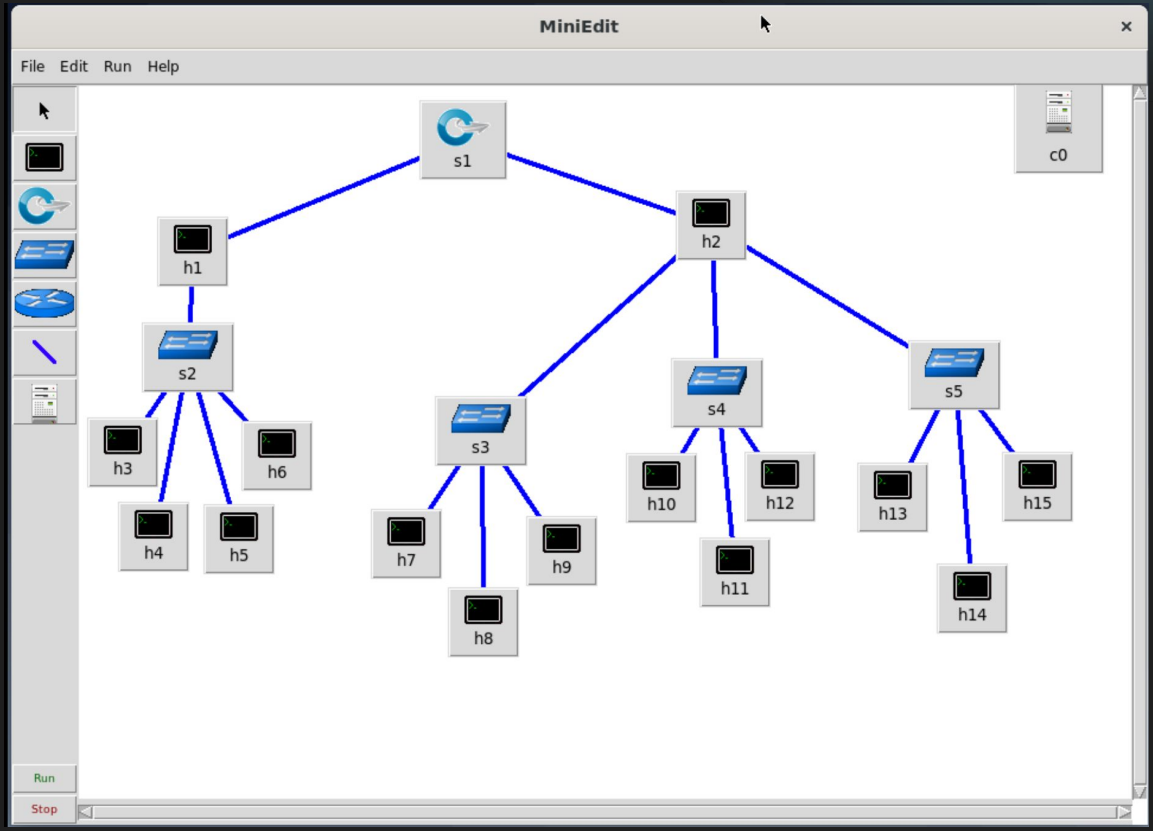
Sprint 1 goals

- Research what the average bandwidth, jitter, packet loss, and latency is for rural, urban, and hospital networks
- Create topologies on miniedit and export the python code
- Create topologies for rural, urban, and hospital networks

Diagram of rural hospital to urban hospital



Topology using miniedit



Demo of rural-to-urban hospital network

All hosts can ping

Some packet loss due to default values

```
mininet> pingall
*** Ping: testing ping reachability
rR -> uR rH1 rH2 rH3 rH4 aH1 X aH3 cH1 cH2 cH3 iH1 iH2 iH3
uR -> rR rH1 rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH2 iH3
rH1 -> rR uR rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH2 X
rH2 -> rR uR rH1 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 X iH3
rH3 -> rR uR rH1 rH2 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH2 iH3
rH4 -> rR uR rH1 rH2 rH3 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH2 iH3
aH1 -> rR uR rH1 rH2 rH3 rH4 aH2 aH3 cH1 cH2 cH3 iH1 iH2 iH3
aH2 -> rR uR rH1 rH2 rH3 rH4 aH1 aH3 cH1 cH2 cH3 iH1 iH2 iH3
aH3 -> rR uR X rH2 X rH4 X aH2 cH1 cH2 cH3 iH1 iH2 iH3
cH1 -> rR uR rH1 X rH3 rH4 aH1 aH2 aH3 cH2 cH3 iH1 iH2 iH3
cH2 -> rR uR rH1 rH2 rH3 X aH1 aH2 aH3 cH1 cH3 iH1 iH2 iH3
cH3 -> rR uR rH1 rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 iH1 iH2 iH3
iH1 -> rR uR rH1 rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH2 iH3
iH2 -> rR uR rH1 rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH3
iH3 -> rR uR rH1 rH2 rH3 rH4 aH1 aH2 aH3 cH1 cH2 cH3 iH1 iH2
*** Results: 3% dropped (202/210 received)
```

Demo of rural-to-urban hospital network

```
mininet> rH1 ping -c 3 rH2
PING 10.10.10.3 (10.10.10.3) 56(84) bytes of data.
64 bytes from 10.10.10.3: icmp_seq=1 ttl=64 time=14.3 ms
64 bytes from 10.10.10.3: icmp_seq=2 ttl=64 time=13.1 ms
64 bytes from 10.10.10.3: icmp_seq=3 ttl=64 time=6.47 ms

--- 10.10.10.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 6.472/11.270/14.285/3.429 ms
```

Rural host one pings rural host two

Rural host one pings urban admin host one

```
mininet> rH1 ping -c 3 aH1
PING 10.20.10.2 (10.20.10.2) 56(84) bytes of data.
64 bytes from 10.20.10.2: icmp_seq=1 ttl=62 time=151 ms
64 bytes from 10.20.10.2: icmp_seq=2 ttl=62 time=191 ms
64 bytes from 10.20.10.2: icmp_seq=3 ttl=62 time=181 ms

--- 10.20.10.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 151.238/174.420/191.027/16.896 ms
```

Latency is shown in every ping

Demo of rural-to-urban hospital network

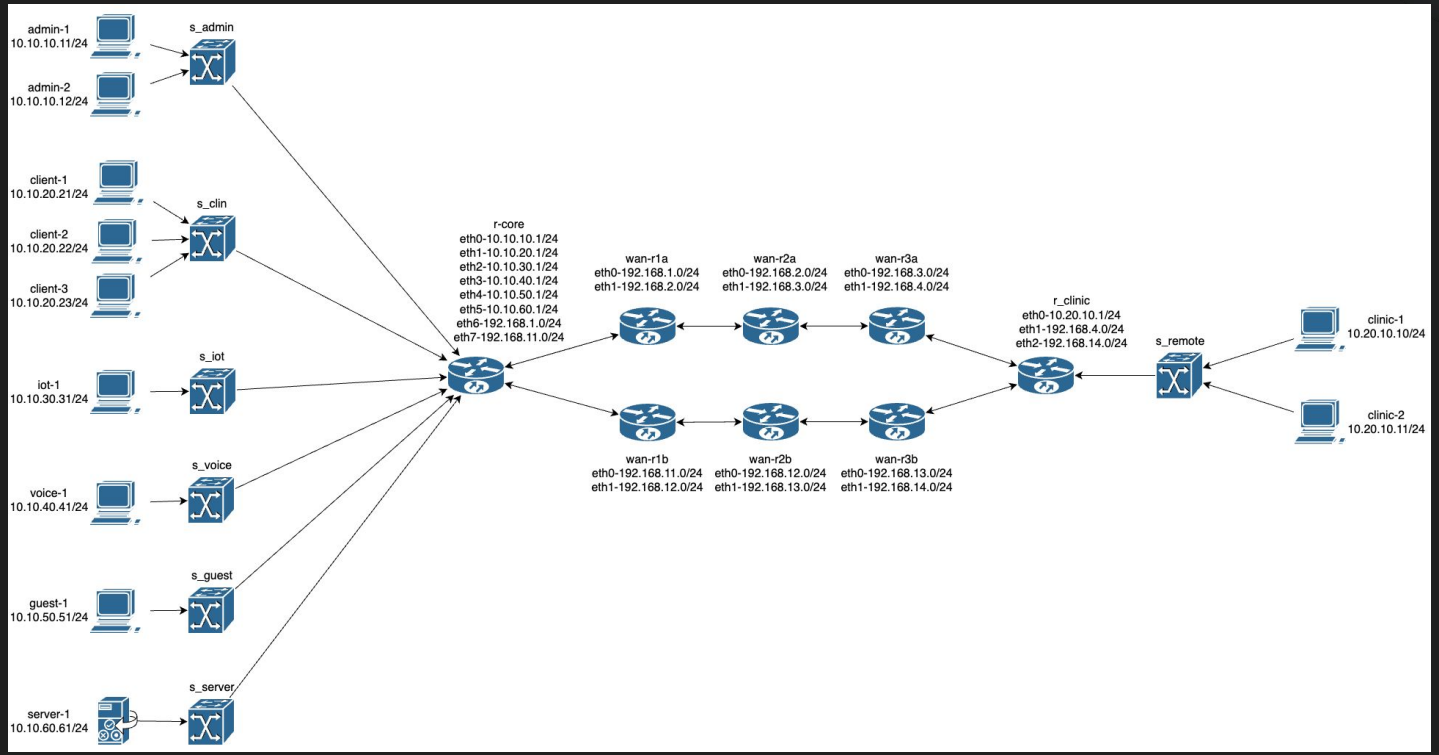
```
mininet> rH1 iperf -u -s &
mininet> rH2 iperf -u -c rH1 -b 10M
------------------------------------------------------------
Client connecting to 10.10.10.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 1121.52 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  3] local 10.10.10.3 port 53632 connected with 10.10.10.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0-10.0 sec  12.5 MBytes  10.5 Mbits/sec
[  3] Sent 8917 datagrams
[  3] Server Report:
[  3]  0.0-10.0 sec  12.4 MBytes  10.4 Mbits/sec   2.303 ms   54/ 8917 (0.61%)
[  3] 0.0000-10.0009 sec  4679 datagrams received out-of-order
```

```
mininet> rH2 iperf -u -c rH1 -b 5M
------------------------------------------------------------
Client connecting to 10.10.10.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 2243.04 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  3] local 10.10.10.3 port 58886 connected with 10.10.10.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0-10.0 sec  6.25 MBytes  5.24 Mbits/sec
[  3] Sent 4459 datagrams
[  3] Server Report:
[  3]  0.0-10.0 sec  6.20 MBytes  5.20 Mbits/sec   3.085 ms   34/ 4459 (0.76%)
[  3] 0.0000-9.9995 sec  1396 datagrams received out-of-order
mininet>
```

Adjusting bandwidth through CLI

Packet loss and jitter shown

Urban-To-Urban Network



Hospital Lan Demo Connection

"Node: admin-1"

```
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet:/home/mininet/mininet/examples# ping -c 3 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=21.0 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=9.50 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=9.30 ms

--- 10.10.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 9.300/13.268/21.009/5.474 ms
root@mininet:/home/mininet/mininet/examples# ip route
default via 10.10.10.1 dev admin-1-eth0
10.10.10.0/24 dev admin-1-eth0 proto kernel scope link src 10.10.10.11
root@mininet:/home/mininet/mininet/examples# ping -c 1 10.10.20.21
PING 10.10.20.21 (10.10.20.21) 56(84) bytes of data.
64 bytes from 10.10.20.21: icmp_seq=1 ttl=63 time=33.3 ms

--- 10.10.20.21 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 33.290/33.290/33.290/0.000 ms
root@mininet:/home/mininet/mininet/examples#
```

Admin-1 Ping to Router
Admin-1 Ping to Client-1

"Node: r-core"

```
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

r-core-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.10.1  netmask 255.255.255.0  broadcast 10.10.10.255
        inet6 fe80::54fc:46ff:fec4:d6e2  prefixlen 64  scopeid 0x20<link>
        ether 56:fc:46:c4:d6:e2  txqueuelen 1000  (Ethernet)
        RX packets 66  bytes 6112 (6.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26  bytes 1944 (1.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

r-core-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.20.1  netmask 255.255.255.0  broadcast 10.10.20.255
        inet6 fe80::1cf2:c2ff:fec0:f2bb  prefixlen 64  scopeid 0x20<link>
        ether 1e:f2:c2:c0:f2:bb  txqueuelen 1000  (Ethernet)
        RX packets 68  bytes 6272 (6.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 16  bytes 1188 (1.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

r-core-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.30.1  netmask 255.255.255.0  broadcast 10.10.30.255
```

Router Subnet Interfaces

Clinic Lan Demo Connection

"Node: clinic-1"

```
        RX packets 54  bytes 5245 (5.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 14  bytes 1076 (1.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet:/home/mininet/mininet/examples# ping -c 3 10.20.10.1
PING 10.20.10.1 (10.20.10.1) 56(84) bytes of data.
64 bytes from 10.20.10.1: icmp_seq=1 ttl=64 time=21.6 ms
64 bytes from 10.20.10.1: icmp_seq=2 ttl=64 time=9.48 ms
64 bytes from 10.20.10.1: icmp_seq=3 ttl=64 time=8.83 ms

--- 10.20.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 8.833/13.314/21.633/5.887 ms
root@mininet:/home/mininet/mininet/examples#
```

Clinic-1 Ping Router

"Node: clinic-2"

```
        RX packets 55  bytes 5335 (5.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 13  bytes 1006 (1.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet:/home/mininet/mininet/examples# ping -c 3 10.20.10.10
PING 10.20.10.10 (10.20.10.10) 56(84) bytes of data.
64 bytes from 10.20.10.10: icmp_seq=1 ttl=64 time=21.3 ms
64 bytes from 10.20.10.10: icmp_seq=2 ttl=64 time=9.45 ms
64 bytes from 10.20.10.10: icmp_seq=3 ttl=64 time=8.88 ms

--- 10.20.10.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 8.883/13.207/21.292/5.721 ms
root@mininet:/home/mininet/mininet/examples#
```

Clinic-2 Ping Clinic 1

Hospital To Clinic WAN Demo Connection

"Node: admin-1"

```
root@mininet:/home/mininet/mininet/examples# ping -c 3 10.20.10.10
PING 10.20.10.10 (10.20.10.10) 56(84) bytes of data.
64 bytes from 10.20.10.10: icmp_seq=1 ttl=59 time=190 ms
64 bytes from 10.20.10.10: icmp_seq=2 ttl=59 time=184 ms
64 bytes from 10.20.10.10: icmp_seq=3 ttl=59 time=185 ms

--- 10.20.10.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 184.044/186.234/189.660/2.453 ms
root@mininet:/home/mininet/mininet/examples# ping -c 3 10.20.10.11
PING 10.20.10.11 (10.20.10.11) 56(84) bytes of data.
64 bytes from 10.20.10.11: icmp_seq=1 ttl=59 time=208 ms
64 bytes from 10.20.10.11: icmp_seq=2 ttl=59 time=184 ms
64 bytes from 10.20.10.11: icmp_seq=3 ttl=59 time=183 ms

--- 10.20.10.11 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 182.958/191.882/208.359/11.664 ms
root@mininet:/home/mininet/mininet/examples# ping -c 3 10.20.10.1
PING 10.20.10.1 (10.20.10.1) 56(84) bytes of data.
64 bytes from 10.20.10.1: icmp_seq=1 ttl=60 time=177 ms
64 bytes from 10.20.10.1: icmp_seq=2 ttl=60 time=175 ms
64 bytes from 10.20.10.1: icmp_seq=3 ttl=60 time=176 ms

--- 10.20.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 174.624/175.876/176.752/0.908 ms
root@mininet:/home/mininet/mininet/examples#
```

Admin-1 Ping Clinic-1, Clinic 2, Clinic Router

"Node: clinic-1"

```
root@mininet:/home/mininet/mininet/examples# ping -c 3 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=60 time=180 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=60 time=176 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=60 time=175 ms

--- 10.10.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 175.359/177.303/180.214/2.096 ms
root@mininet:/home/mininet/mininet/examples# ping -c 3 10.10.20.21
PING 10.10.20.21 (10.10.20.21) 56(84) bytes of data.
64 bytes from 10.10.20.21: icmp_seq=1 ttl=59 time=189 ms
64 bytes from 10.10.20.21: icmp_seq=2 ttl=59 time=186 ms
64 bytes from 10.10.20.21: icmp_seq=3 ttl=59 time=186 ms

--- 10.10.20.21 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 185.713/186.857/189.070/1.564 ms
```

Clinic-1 Ping Admin-1, Client-1

Other Contributions - Fernando

- Working, mainly, on the UI portion of the project
- Work:
  - Creating the scene for topologies to be rendered and interactable
  - Creating “properties” panel that appears when configuring clients
  - Research how to apply scripts from GUI
  - Research link of GUI and Mininet CLI environment
- Goal:
  - Import topologies and show them on screen
  - Change link characteristics from the GUI after topology is loaded
  - Apply attacks from GUI by choosing a client
  - Provide both CLI and GUI interface for flexibility

Other Contributions - Dylan & Selim

Finding a way to attack our network - Done

- Mininet-sec - For emulation platform for studying and experimenting cybersecurity tools in programmable networks.

Testing attacks on our emulated network - MVP

- Was able to test the attacks on example python scripts, have not tried to test it on our own network topologies

Example 1 :

```
selim@selim-VirtualBox:~/mininet-sec$ sudo python3 topology-test.py
[sudo] password for selim:
*** Adding controller
*** Add switches
*** Add router
*** Add hosts
*** Add links
*** Starting network
*** Configuring hosts
r3 h1 h2 h3 h4
*** Starting controllers
*** Starting switches
*** Configuring router interfaces
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
r3 -> h1 h2 h3 h4
h1 -> r3 h2 h3 h4
h2 -> r3 h1 h3 h4
h3 -> r3 h1 h2 h4
h4 -> r3 h1 h2 h3
*** Results: 0% dropped (20/20 received)
mininet> h1 ping -c 2 192.168.11.1
PING 192.168.11.1 (192.168.11.1) 56(84) bytes of data.
64 bytes from 192.168.11.1: icmp_seq=1 ttl=63 time=0.711 ms
64 bytes from 192.168.11.1: icmp_seq=2 ttl=63 time=0.166 ms

--- 192.168.11.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.166/0.438/0.711/0.272 ms
mininet> h3 python3 -m http.server 8080 &
mininet> h3 ss -tulpn
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
Netid  State   Recv-Q  Send-Q   Local Address:Port   Peer Address:Port Process
tcp    LISTEN  0       5              0.0.0.0:8080        0.0.0.0:*     users:(("python3",pid=3852,fd=3))
```

Next Steps

Sprint 2 - 03/12

Create a basic UI that imports and shows topologies
Reflect modifications from UI to Mininet

Sprint 3 - 03/26

- Incorporate a traffic generator into UI
- Create network attack scripts and incorporate into UI
  - DDoS, arp spoofing, eavesdropping, etc.

Sprint 4 - 04/09

- Test all attacks on created networks
- Incorporate attack scripts to UI
- Track findings

Thank you!